

Supplementary Material for “Leakage-Safe Evaluation of Machine-Learning Models for Mental-Health Prediction from Actigraphy: Protocol, Benchmark and Appraisal of Published Claims”

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I. TABLE S1: TRIPOD+AI CHECKLIST

TABLE S1: TRIPOD+AI checklist (Collins et al. 2024) for this study; item wording abridged, numbering as in the companion review’s extraction workbook.

Item	Requirement	Where reported
1	Title identifies development/validation of a prediction model, target population, outcome	Title
2	Abstract reports objective, data, population, outcome, predictors, sample size, model, performance	Abstract
3a	Background and rationale	Introduction para 1-2
3b	Objectives incl. development/validation	Introduction, final para (E1-E3)
4a	Data source described	Methods 2.1
4b	Study dates	Methods 2.1
5a	Setting, eligibility, recruitment	Methods 2.1; Table 1
5b	Treatments received	Methods 2.1 (not used)
6	Outcome definition and timing	Methods 2.2
7	Predictors defined incl. measurement	Methods 2.3; Supplement S2
8	Sample size justified, EPV stated	Methods 2.4; Table 3 (EPV)
9	Missing data handling	Methods 2.5
10a	Pre-processing	Methods 2.5-2.6
10b	Model types and rationale	Methods 2.6
10c	Hyperparameter tuning	Methods 2.6 (fixed a priori)
10d	Internal validation (CV scheme, repeats, seeds)	Methods 2.7
10e	External validation method	Methods 2.8
11	Class imbalance handling; resampling inside folds	Methods 2.6
12	Performance measures incl. discrimination AND calibration	Methods 2.9
13	Model updating / recalibration	Methods 2.10
14	Fairness / subgroup performance	Methods 2.11; Results 3.5
15	Interpretability methods	Discussion (not an aim)
16	Funding	Declarations
17	Conflicts of interest	Declarations
18	Protocol / registration	Declarations (OSF)
19	Data availability	Data availability
20	Code availability	Code availability
21	Participant flow	Results 3.1
22	Participant characteristics incl. prevalence	Table 1
23	Full model specification	Methods 2.6; repository
24	Performance with uncertainty intervals	Tables 2-4
25	Subgroup / fairness results	Results 3.5; Table 5
26	Limitations	Discussion 4.3
27	Interpretation and implications	Discussion 4.4

II. TABLE S2: FEATURE DEFINITIONS

TABLE S2: Predictors computed for each valid calendar day of minute-level activity counts.

Feature	Definition
mean, sd, cv	Mean, standard deviation and coefficient of variation of counts over valid minutes
median, p10, p90, max	Median, 10th and 90th percentiles, maximum
skew, kurtosis	Sample skewness and excess kurtosis
prop_zero	Proportion of valid minutes with zero counts (inactivity)

Feature	Definition
mean_log1p, sd_log1p	Mean and SD of $\log(1 + x)$
day_mean, night_mean	Mean counts 08:00–22:00 and outside that interval
night_day_ratio	night_mean / day_mean
night_prop_zero	Proportion of zero minutes at night
M10, L5	Mean of the most active 10 h and least active 5 h of the hourly profile (circular)
relative_amplitude	$(M10 - L5)/(M10 + L5)$
IV	Intradaily variability of the hourly profile (Van Someren et al.)
m10_onset_sin, m10_onset_cos	Sine and cosine of the hour at which the M10 window starts (phase proxy)
acf_5, acf_60	Autocorrelation of the minute series at lags 5 and 60 min
psd_mean	$\log(1 + \text{mean power})$ of the within-day periodogram (excluding DC)
dominant_period_min	Period (min) of the dominant within-day spectral peak
n_transitions	Number of rest/activity transitions (zero to non-zero or back)
missing_frac	Fraction of the 1440 minutes without a valid count

III. TABLES S3 AND S4: IDENTIFICATION SOURCES AND STUDY LIST

TABLE S3
SUPPLEMENTARY TABLE S3. IDENTIFICATION SOURCES FOR PART I (SEPTEMBER 2026).

Source	Query / procedure
Google Scholar	“Depresjon” OR “Psykose” OR “Hyperaktiv” OR “OBF-Psychiatric”; “Cited by” on
Semantic Scholar	citation graph of the four dataset papers
PubMed / PMC	Depresjon[tiab] OR Psykose[tiab] OR Hyperaktiv[tiab] OR “OBF-Psychiatric”[tiab]
arXiv / medRxiv / TechRxiv	full-text search for the four dataset names
Publisher sites (Springer, Elsevier, MDPI, ACM, IEEE, JMIR)	full-text search for the four dataset names

TABLE S4: Supplementary Table S4. Study sources and extraction basis.

ID	Source	Extraction basis
S01	doi:10.1145/3204949.3208125	full text
S02	Feature extraction in motor activity signal (Diagnostics 2019)	abstract/metadata
S03	doi:10.1145/3347444.3356238	abstract/metadata
S04	doi:10.1371/journal.pone.0231995	abstract/metadata
S05	doi:10.1109/CBMS49503.2020.00064	full text
S06	doi:10.1145/3458305.3478454	full text
S07	PMC9495338	abstract/metadata
S08	PMC9318635	abstract/metadata
S09	ResearchGate 362753700	abstract/metadata
S10	arXiv:2303.07847	abstract/metadata
S11	arXiv:2310.09277	abstract/metadata
S12	PMC10377958	abstract/metadata
S13	SciELO S0188-95322023000400038	abstract/metadata
S14	doi:10.1007/s11042-023-17119-6	abstract/metadata
S15	doi:10.1007/s00500-024-09832-7	abstract/metadata
S16	doi:10.1016/j.bspc.2024.105907 (verify)	abstract/metadata
S17	doi:10.1007/s11042-024-19631-9	abstract/metadata
S18	doi:10.1007/s41870-024-01895-x	abstract/metadata
S19	doi:10.2196/72038	abstract/metadata
S20	doi:10.1038/s41598-025-07245-6	abstract/metadata
S21	PMID 40588896	abstract/metadata
S22	doi:10.1007/978-3-032-05507-1_16	abstract/metadata
S23	doi:10.36227/techrxiv.176403977.77890479	abstract/metadata
S24	doi:10.1038/s41597-025-04384-3	full text
S25	doi:10.1101/2025.08.26.25332257	abstract/metadata
S26	arXiv:2512.00103	abstract/metadata
S27	doi:10.1007/s44163-026-01574-9	abstract/metadata
S28	PMC9312518	abstract/metadata

IV. TABLE S5: RUN MANIFESTS AND PREDICTION FILES

Every experiment run writes `results/real/<experiment>/manifest.json` (timestamp, git commit and dirty flag, configuration hash and configuration, interpreter and platform, package versions, SHA-256 of every raw file, synthetic flag, experiment name, rows written) and per-window out-of-fold predictions `results/real/<experiment>/predictions/<cohort>.<model>.<splitter>.csv` (seed, fold, window_id, subject_id, cohort, y, p). Tables are re-generated by `python-mmlmhretable`, `scripts/part1_figure.py`, `scripts/part2_extra_figures.py` and `scripts/fill_e*_paragraph.py`.

TABLE S5

PART I: ELIGIBLE PUBLISHED SUPERVISED MODELS ON THE FOUR COHORTS (SOURCES IN SUPPLEMENTARY TABLE S4). ^aEXTRACTED FROM THE ABSTRACT AND METADATA (FULL TEXT NOT ACCESSIBLE AT EXTRACTION); N.R. = NOT REPORTED THERE. SPLIT UNIT: PARTICIPANT = LEAVE-ONE-PARTICIPANT-OUT OR GROUPED FOLDS; DAY/WINDOW = FOLDS OVER DAYS OR WINDOWS; UNSTATED = NOT STATED IN THE AVAILABLE TEXT.

ID	First author	Year	Cohorts	Best model	Split unit	Best reported	Calib.	Ext. val.	Code
S01	Garcia-Ceja	2018	DEPRESJON	Random forest / DNN	participant (leave-one-patient-out)	F1 0.73	no	no	yes
S02 ^a	Zanella-Calzada	2019	DEPRESJON	Random forest	day/window	accuracy 0.89	no	no	no
S03 ^a	Frogner	2019	DEPRESJON	1D-CNN	participant (leave-one-participant-out)	F1 n.r.	no	no	no
S04 ^a	Jakobsen	2020	DEPRESJON	Random forest / DNN / CNN	participant (leave-one-out)	accuracy n.r.	no	no	no
S05	Jakobsen	2020	PSYKOSE	Classical ML	participant (leave-one-patient-out)	F1 n.r.	no	no	yes
S06	Hicks	2021	HYPERAKTIV	Classical ML	unstated (10-fold on features)	accuracy n.r.	no	no	yes
S07 ^a	Rodriguez-Ruiz	2022	DEPRESJON	2D-CNN	day/window	accuracy 0.767	no	no	no
S08 ^a	Rodriguez-Ruiz	2022	DEPRESJON+PSYKOSE	CNN	day/window	accuracy n.r.	no	no	no
S09 ^a	Kumar	2022	DEPRESJON	ML on autoregressive features	unstated	accuracy 0.78	no	no	no
S10 ^a	Ahmed (arXiv)	2023	DEPRESJON	Transfer learning ensemble	modified leave-one-out	accuracy 0.96	no	claims independent set	no
S11 ^a	Hybrid RF-ANN (arXiv)	2023	DEPRESJON	RF-ANN ensemble	unstated	accuracy n.r.	no	no	no
S12 ^a	UMAP study	2023	DEPRESJON	ML + UMAP	unstated	accuracy n.r.	no	no	no
S13 ^a	Genetic-algorithm feature selection	2023	DEPRESJON	GA feature selection + classifier	day/window (intervals)	accuracy n.r.	no	no	no
S14 ^a	Attention CNN (IoMT)	2023	PSYKOSE	Deep CNN + attention	day/window	accuracy n.r.	no	no	no
S15 ^a	Multi-branch DL	2024	PSYKOSE	Multi-branch deep network	day/window	accuracy 0.92	no	no	no
S16 ^a	DL + XAI	2024	PSYKOSE (+DEPRESJON)	Deep network + XAI	day/window (24-h segments)	accuracy 0.94	no	no	no
S17 ^a	Hopping-mean augmentation	2024	DEPRESJON	Augmented ML	day/window	accuracy n.r.	no	no	no
S18 ^a	IoMT dimensionality reduction	2024	HYPERAKTIV	10 classifiers on reduced features	unstated	accuracy n.r.	no	no	no
S19 ^a	Explainable XGBoost	2025	DEPRESJON	XGBoost + ADASYN	unstated in abstract	accuracy 0.849	no	no	no
S20 ^a	Improved CNN framework	2025	DEPRESJON+PSYKOSE	Improved CNN	day/window	accuracy n.r.	no	no	no
S21 ^a	Enhanced ADHD ML	2025	HYPERAKTIV/OBF (45 participants)	SVM	unstated	F1 0.779	no	no	no
S22 ^a	HYPERAKTIV ML study	2025	HYPERAKTIV	RF / LightGBM / SVM	unstated	accuracy n.r.	no	no	no
S23 ^a	Multimodal digital biomarkers	2025	HYPERAKTIV	Bernoulli NB (actigraphy+HRV)	unstated	F1 0.79	no	no	no
S24	OBF-Psychiatric descriptor	2025	OBF-Psychiatric	Classical ML baselines	participant	accuracy n.r.	no	no	yes
S25 ^a	Data-uncertainty case study	2025	OBF-Psychiatric (ADHD)	QC + ML	participant	nan no difference after QC	no	no	no
S26 ^a	VIT / CoAtNet on images	2025	DEPRESJON+PSYKOSE	CoAtNet-Tiny	participant (three-fold subject-wise)	accuracy n.r.	no	no	no
S27 ^a	Time-segmentation evaluation	2026	DEPRESJON	ML on day segments	n.r.	accuracy n.r.	no	no	no
S28 ^a	Accurate ADHD identification	2022	HYPERAKTIV	ML on real-time activity	unstated	accuracy n.r.	no	no	no

V. TABLE S6: 1D-CNN ARM

TABLE S6

E1, WINDOW-LEVEL AUROC UNDER SUBJECT-WISE VERSUS RECORD-WISE FIVE-FOLD CROSS-VALIDATION. ESTIMATES ON SEED-AVERAGED OUT-OF-FOLD PREDICTIONS WITH SUBJECT-LEVEL BCA 95% BOOTSTRAP INTERVALS; INFLATION IS THE PAIRED RECORD-WISE MINUS SUBJECT-WISE DIFFERENCE.

Cohort	Model	AUROC subject-wise	AUROC record-wise	Inflation (paired)
depresjon	1D-CNN	0.696 [0.576, 0.795]	0.729 [0.671, 0.784]	0.033 [-0.063, 0.122]
psykose	1D-CNN	0.849 [0.744, 0.910]	0.892 [0.836, 0.935]	0.043 [-0.000, 0.095]
hyperaktiv	1D-CNN	0.459 [0.356, 0.588]	0.513 [0.447, 0.578]	0.054 [-0.079, 0.189]

TABLE S7

E1, PARTICIPANT-LEVEL AUROC UNDER SUBJECT-WISE VERSUS RECORD-WISE FIVE-FOLD CROSS-VALIDATION. ESTIMATES ON SEED-AVERAGED OUT-OF-FOLD PREDICTIONS WITH SUBJECT-LEVEL BCA 95% BOOTSTRAP INTERVALS; INFLATION IS THE PAIRED RECORD-WISE MINUS SUBJECT-WISE DIFFERENCE.

Cohort	Model	AUROC subject-wise	AUROC record-wise	Inflation (paired)
depresjon	1D-CNN	0.738 [0.634, 0.844]	0.912 [0.837, 0.959]	0.174 [0.067, 0.270]
psykose	1D-CNN	0.922 [0.852, 0.976]	0.986 [0.963, 1.000]	0.064 [0.016, 0.111]
hyperaktiv	1D-CNN	0.455 [0.371, 0.563]	0.529 [0.432, 0.622]	0.074 [-0.060, 0.208]

VI. TABLE S7: CALIBRATION OF EVERY MODEL (E3)

VII. TABLE S8: ABLATION (E6)

VIII. TABLE S9: SEX-STRATIFIED PERFORMANCE (E4)

IX. TABLE S10: FULL WINDOW-LEVEL METRICS (E1)

X. FIGURES S1 TO S3: ROC AND RELIABILITY CURVES

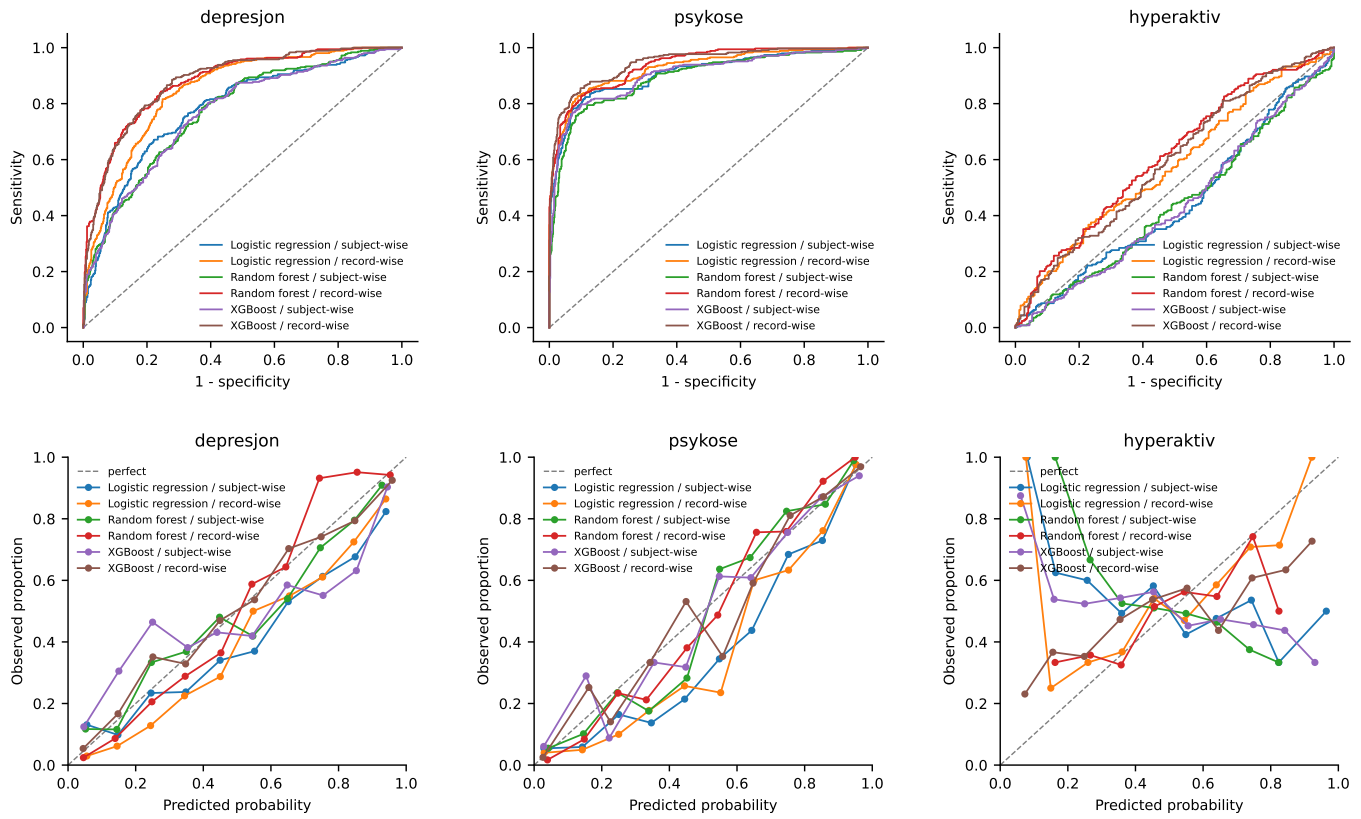


Fig. S1. E1. ROC (top) and reliability (bottom) curves per cohort under subject-wise and record-wise cross-validation.

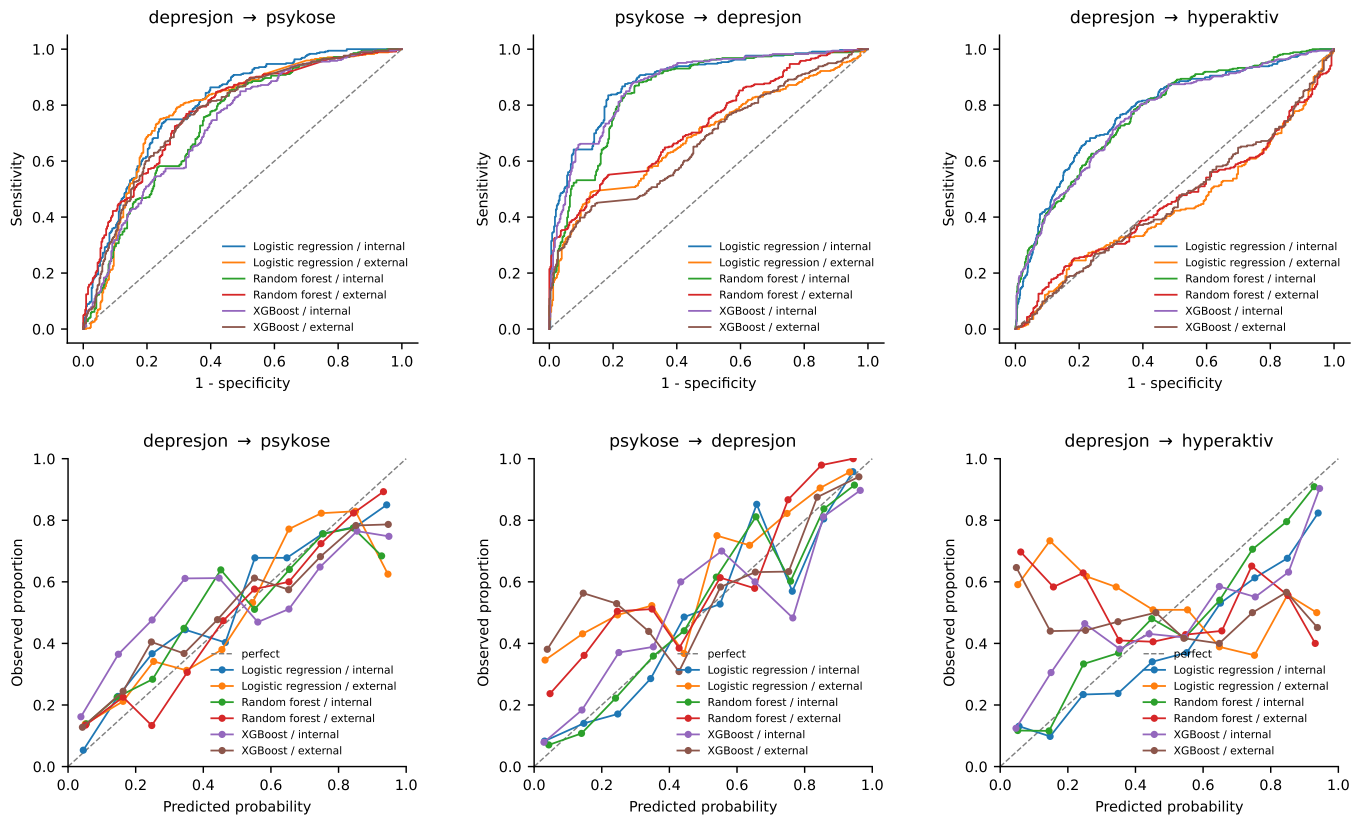


Fig. S2. E2. ROC (top) and reliability (bottom) curves of internal and external predictions per cohort pair.

TABLE S8

E3: CALIBRATION ALONGSIDE DISCRIMINATION FOR EVERY MODEL IN E1 (SUBJECT-WISE ARM) AND E2, PLUS LOGISTIC RECALIBRATION OF EXTERNAL PREDICTIONS.

Exp.	Run	AUROC	Brier	Cal. slope	Cal. intercept	ECE
E1	depresjon.logreg.subject_wise	0.779	0.187	0.69	-0.51	0.102
E1	depresjon.rf.subject_wise	0.770	0.181	0.77	0.01	0.056
E1	depresjon.xgboost.subject_wise	0.764	0.192	0.53	0.27	0.105
E1	hyperaktiv.logreg.subject_wise	0.449	0.279	-0.28	-0.01	0.161
E1	hyperaktiv.rf.subject_wise	0.443	0.281	-0.45	0.00	0.137
E1	hyperaktiv.xgboost.subject_wise	0.442	0.313	-0.23	0.00	0.221
E1	psykose.logreg.subject_wise	0.907	0.112	0.95	-0.59	0.088
E1	psykose.rf.subject_wise	0.893	0.115	1.03	-0.12	0.052
E1	psykose.xgboost.subject_wise	0.903	0.103	0.78	0.14	0.052
E2	depresjon-to-hyperaktiv.logreg.external	0.442	0.350	-0.16	-0.42	0.291
E2	depresjon-to-hyperaktiv.rf.external	0.461	0.317	-0.15	0.33	0.214
E2	depresjon-to-hyperaktiv.xgboost.external	0.459	0.363	-0.09	0.47	0.292
E2	depresjon-to-psykose.logreg.external	0.779	0.193	0.50	0.06	0.079
E2	depresjon-to-psykose.rf.external	0.776	0.193	0.81	-0.02	0.044
E2	depresjon-to-psykose.xgboost.external	0.769	0.202	0.56	-0.01	0.085
E2	depresjon.logreg.internal	0.779	0.187	0.69	-0.51	0.102
E2	depresjon.rf.internal	0.770	0.181	0.77	0.01	0.056
E2	depresjon.xgboost.internal	0.764	0.192	0.53	0.27	0.105
E2	psykose-to-depresjon.logreg.external	0.689	0.262	0.27	1.20	0.202
E2	psykose-to-depresjon.rf.external	0.728	0.225	0.63	0.74	0.149
E2	psykose-to-depresjon.xgboost.external	0.671	0.284	0.31	1.43	0.229
E2	psykose.logreg.internal	0.885	0.137	0.77	-0.10	0.061
E2	psykose.rf.internal	0.854	0.151	0.82	-0.08	0.055
E2	psykose.xgboost.internal	0.877	0.148	0.62	-0.16	0.083
E2	depresjon-to-hyperaktiv.logreg.external+recal	0.442	0.311	-0.23	0.22	0.223
E2	depresjon-to-hyperaktiv.rf.external+recal	0.461	0.301	-0.20	0.37	0.176
E2	depresjon-to-hyperaktiv.xgboost.external+recal	0.459	0.311	-0.17	0.36	0.194
E2	depresjon-to-psykose.logreg.external+recal	0.779	0.211	0.72	0.56	0.139
E2	depresjon-to-psykose.rf.external+recal	0.776	0.193	1.05	0.11	0.041
E2	depresjon-to-psykose.xgboost.external+recal	0.769	0.196	1.05	0.11	0.052
E2	psykose-to-depresjon.logreg.external+recal	0.689	0.254	0.35	0.99	0.185
E2	psykose-to-depresjon.rf.external+recal	0.728	0.221	0.77	0.67	0.129
E2	psykose-to-depresjon.xgboost.external+recal	0.671	0.258	0.49	0.93	0.200

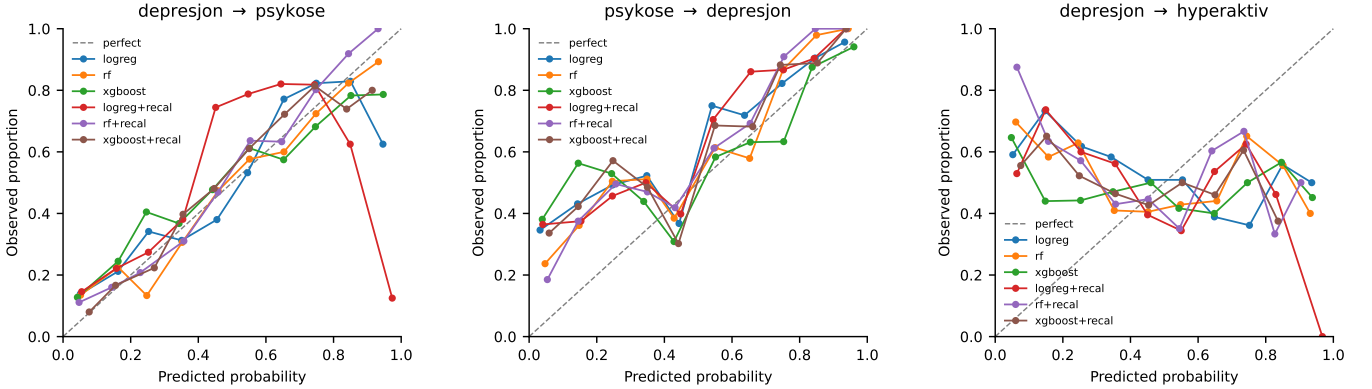


Fig. S3. E3. Reliability of external predictions before and after logistic recalibration fitted on the training cohort.

TABLE S9

E6: ABLATION UNDER SUBJECT-WISE CV. FEATURE-GROUP ABLATIONS (DROP-ONE AND ONLY-ONE), SMOTE APPLIED INSIDE TRAINING FOLDS, AND LEAVE-ONE-SUBJECT-OUT (LOSO) IN PLACE OF FIVE-FOLD CV. WINDOW-LEVEL ESTIMATES WITH SUBJECT-LEVEL BCA INTERVALS.

Cohort	Model	Variant	k	AUROC [95% CI]	Subject AUROC	Brier	Cal. slope	ECE
depresjon	Logistic regression	all_features	28	0.785 [0.694, 0.853]	0.902	0.187	0.71	0.103
depresjon	Logistic regression	drop_distributional	16	0.758 [0.649, 0.837]	0.890	0.198	0.69	0.119
depresjon	Logistic regression	only_distributional	12	0.755 [0.652, 0.836]	0.838	0.200	0.79	0.115
depresjon	Logistic regression	drop_day_night	24	0.781 [0.687, 0.850]	0.893	0.188	0.74	0.108
depresjon	Logistic regression	only_day_night	4	0.458 [0.358, 0.541]	0.577	0.257	-0.07	0.149
depresjon	Logistic regression	drop_circadian	22	0.797 [0.708, 0.863]	0.906	0.181	0.78	0.103
depresjon	Logistic regression	only_circadian	6	0.640 [0.533, 0.722]	0.783	0.239	0.59	0.152
depresjon	Logistic regression	drop_temporal_spectral	22	0.771 [0.678, 0.840]	0.890	0.193	0.73	0.102
depresjon	Logistic regression	only_temporal_spectral	6	0.740 [0.551, 0.829]	0.874	0.211	0.85	0.178
depresjon	Logistic regression	all_features+smote_in_fold	28	0.786 [0.697, 0.852]	0.899	0.186	0.68	0.097
depresjon	Logistic regression	all_features+loso	28	0.773 [0.660, 0.849]	0.890	0.191	0.63	0.111
depresjon	XGBoost	all_features	28	0.768 [0.685, 0.837]	0.871	0.192	0.53	0.101
depresjon	XGBoost	drop_distributional	16	0.788 [0.711, 0.852]	0.905	0.181	0.59	0.093
depresjon	XGBoost	only_distributional	12	0.739 [0.640, 0.812]	0.844	0.196	0.54	0.090
depresjon	XGBoost	drop_day_night	24	0.769 [0.685, 0.836]	0.878	0.190	0.53	0.093
depresjon	XGBoost	only_day_night	4	0.769 [0.682, 0.841]	0.875	0.185	0.63	0.062
depresjon	XGBoost	drop_circadian	22	0.787 [0.707, 0.851]	0.912	0.181	0.58	0.090
depresjon	XGBoost	only_circadian	6	0.711 [0.606, 0.796]	0.817	0.203	0.52	0.090
depresjon	XGBoost	drop_temporal_spectral	22	0.740 [0.642, 0.817]	0.825	0.204	0.49	0.117
depresjon	XGBoost	only_temporal_spectral	6	0.802 [0.737, 0.859]	0.947	0.168	0.68	0.053
depresjon	XGBoost	all_features+smote_in_fold	28	0.761 [0.676, 0.835]	0.879	0.196	0.51	0.109
depresjon	XGBoost	all_features+loso	28	0.745 [0.649, 0.820]	0.853	0.205	0.45	0.124
psykose	Logistic regression	all_features	28	0.903 [0.807, 0.937]	0.973	0.115	0.92	0.090
psykose	Logistic regression	drop_distributional	16	0.845 [0.754, 0.892]	0.952	0.159	0.87	0.119
psykose	Logistic regression	only_distributional	12	0.875 [0.785, 0.921]	0.945	0.136	0.95	0.099
psykose	Logistic regression	drop_day_night	24	0.890 [0.825, 0.926]	0.979	0.125	0.94	0.083
psykose	Logistic regression	only_day_night	4	0.703 [0.575, 0.791]	0.852	0.223	0.74	0.195
psykose	Logistic regression	drop_circadian	22	0.890 [0.810, 0.932]	0.967	0.125	0.92	0.092
psykose	Logistic regression	only_circadian	6	0.784 [0.714, 0.839]	0.926	0.192	0.85	0.119
psykose	Logistic regression	drop_temporal_spectral	22	0.900 [0.810, 0.936]	0.969	0.117	0.95	0.095
psykose	Logistic regression	only_temporal_spectral	6	0.733 [0.665, 0.790]	0.862	0.212	0.94	0.146
psykose	Logistic regression	all_features+smote_in_fold	28	0.906 [0.810, 0.939]	0.973	0.113	0.86	0.082
psykose	Logistic regression	all_features+loso	28	0.906 [0.811, 0.939]	0.974	0.114	0.92	0.092
psykose	XGBoost	all_features	28	0.902 [0.806, 0.939]	0.969	0.104	0.77	0.045
psykose	XGBoost	drop_distributional	16	0.901 [0.813, 0.937]	0.969	0.103	0.79	0.064
psykose	XGBoost	only_distributional	12	0.874 [0.768, 0.920]	0.950	0.124	0.74	0.054
psykose	XGBoost	drop_day_night	24	0.887 [0.793, 0.926]	0.956	0.114	0.78	0.061
psykose	XGBoost	only_day_night	4	0.854 [0.739, 0.903]	0.956	0.134	0.73	0.059
psykose	XGBoost	drop_circadian	22	0.897 [0.796, 0.937]	0.967	0.105	0.76	0.052
psykose	XGBoost	only_circadian	6	0.843 [0.763, 0.889]	0.935	0.143	0.77	0.060
psykose	XGBoost	drop_temporal_spectral	22	0.900 [0.807, 0.935]	0.963	0.108	0.77	0.045
psykose	XGBoost	only_temporal_spectral	6	0.854 [0.781, 0.905]	0.969	0.135	0.75	0.046
psykose	XGBoost	all_features+smote_in_fold	28	0.904 [0.811, 0.940]	0.972	0.106	0.76	0.046
psykose	XGBoost	all_features+loso	28	0.899 [0.810, 0.936]	0.970	0.109	0.73	0.051
hyperaktiv	Logistic regression	all_features	28	0.466 [0.383, 0.554]	0.456	0.275	-0.17	0.157
hyperaktiv	Logistic regression	drop_distributional	16	0.472 [0.401, 0.557]	0.459	0.267	-0.15	0.130
hyperaktiv	Logistic regression	only_distributional	12	0.461 [0.364, 0.544]	0.426	0.264	-0.37	0.082
hyperaktiv	Logistic regression	drop_day_night	24	0.476 [0.394, 0.556]	0.455	0.270	-0.08	0.133
hyperaktiv	Logistic regression	only_day_night	4	0.493 [0.376, 0.594]	0.471	0.258	-0.40	0.047
hyperaktiv	Logistic regression	drop_circadian	22	0.418 [0.323, 0.510]	0.391	0.277	-0.52	0.133
hyperaktiv	Logistic regression	only_circadian	6	0.486 [0.398, 0.558]	0.495	0.259	-0.30	0.051
hyperaktiv	Logistic regression	drop_temporal_spectral	22	0.493 [0.406, 0.577]	0.490	0.267	-0.07	0.130
hyperaktiv	Logistic regression	only_temporal_spectral	6	0.519 [0.418, 0.612]	0.505	0.255	0.05	0.028
hyperaktiv	Logistic regression	all_features+smote_in_fold	28	0.466 [0.383, 0.555]	0.453	0.275	-0.17	0.149
hyperaktiv	Logistic regression	all_features+loso	28	0.472 [0.388, 0.553]	0.457	0.275	-0.16	0.146
hyperaktiv	XGBoost	all_features	28	0.452 [0.378, 0.548]	0.444	0.311	-0.18	0.207
hyperaktiv	XGBoost	drop_distributional	16	0.453 [0.383, 0.535]	0.451	0.307	-0.17	0.207
hyperaktiv	XGBoost	only_distributional	12	0.517 [0.442, 0.594]	0.536	0.287	0.03	0.155
hyperaktiv	XGBoost	drop_day_night	24	0.467 [0.398, 0.562]	0.477	0.304	-0.11	0.198
hyperaktiv	XGBoost	only_day_night	4	0.493 [0.424, 0.573]	0.504	0.288	-0.07	0.154
hyperaktiv	XGBoost	drop_circadian	22	0.454 [0.373, 0.540]	0.445	0.306	-0.16	0.210
hyperaktiv	XGBoost	only_circadian	6	0.459 [0.398, 0.524]	0.461	0.295	-0.20	0.171
hyperaktiv	XGBoost	drop_temporal_spectral	22	0.471 [0.401, 0.565]	0.467	0.304	-0.14	0.195
hyperaktiv	XGBoost	only_temporal_spectral	6	0.470 [0.407, 0.538]	0.460	0.292	-0.09	0.174
hyperaktiv	XGBoost	all_features+smote_in_fold	28	0.454 [0.379, 0.543]	0.453	0.310	-0.18	0.208
hyperaktiv	XGBoost	all_features+loso	28	0.424 [0.344, 0.503]	0.399	0.334	-0.25	0.275

TABLE S10

E4: SEX-STRATIFIED PERFORMANCE OF THE SUBJECT-WISE E1 MODELS (WINDOW LEVEL, SUBJECT-LEVEL BOOTSTRAP INTERVALS). SUBGROUPS WITH FEWER THAN SIX PARTICIPANTS OR A SINGLE CLASS ARE OMITTED.

Cohort	Model	Sex	Participants (cases)	AUROC [95\% CI]	Brier	Cal. slope	ECE
depresjon	Logistic regression	female	30 (10)	0.777 [0.649, 0.872]	0.182	0.75	0.120
depresjon	Logistic regression	male	25 (13)	0.776 [0.562, 0.882]	0.192	0.62	0.091
depresjon	Random forest	female	30 (10)	0.763 [0.623, 0.870]	0.171	0.75	0.071
depresjon	Random forest	male	25 (13)	0.777 [0.626, 0.870]	0.193	0.80	0.070
depresjon	XGBoost	female	30 (10)	0.755 [0.620, 0.859]	0.183	0.51	0.109
depresjon	XGBoost	male	25 (13)	0.772 [0.619, 0.868]	0.203	0.55	0.120
hyperaktiv	Logistic regression	female	41 (21)	0.479 [0.370, 0.602]	0.275	-0.10	0.152
hyperaktiv	Logistic regression	male	44 (24)	0.424 [0.321, 0.543]	0.283	-0.49	0.168
hyperaktiv	Random forest	female	41 (21)	0.491 [0.371, 0.611]	0.269	-0.10	0.111
hyperaktiv	Random forest	male	44 (24)	0.398 [0.278, 0.516]	0.291	-0.80	0.160
hyperaktiv	XGBoost	female	41 (21)	0.498 [0.385, 0.605]	0.293	-0.02	0.211
hyperaktiv	XGBoost	male	44 (24)	0.388 [0.278, 0.505]	0.332	-0.44	0.230
psykose	Logistic regression	female	23 (3)	0.971 [0.915, 0.998]	0.096	1.58	0.177
psykose	Logistic regression	male	31 (19)	0.890 [0.771, 0.933]	0.126	0.83	0.081
psykose	Random forest	female	23 (3)	0.966 [0.891, 0.995]	0.075	1.66	0.147
psykose	Random forest	male	31 (19)	0.874 [0.765, 0.930]	0.149	0.90	0.123
psykose	XGBoost	female	23 (3)	0.969 [0.904, 0.997]	0.059	1.16	0.100
psykose	XGBoost	male	31 (19)	0.884 [0.774, 0.933]	0.140	0.71	0.121

TABLE S11

E1: WINDOW-LEVEL DISCRIMINATION AND CALIBRATION FOR EVERY MODEL, COHORT AND SPLITTING DESIGN (PER-SEED MEANS OVER TEN SEEDS).

Cohort	Model	Split	Accuracy	Macro-F1	AUROC	Brier	Cal. slope	Cal. intercept	ECE
depresjon	Majority class	subject-wise	0.653	0.395	0.482	0.227	-4.01	-0.00	0.000
depresjon	Majority class	record-wise	0.653	0.395	0.499	0.227	0.29	0.00	0.000
depresjon	Logistic regression	subject-wise	0.722	0.706	0.765	0.194	0.52	-0.54	0.118
depresjon	Logistic regression	record-wise	0.761	0.750	0.840	0.163	0.90	-0.62	0.095
depresjon	Random forest	subject-wise	0.724	0.679	0.762	0.185	0.69	0.01	0.061
depresjon	Random forest	record-wise	0.806	0.782	0.874	0.136	1.23	-0.11	0.039
depresjon	XGBoost	subject-wise	0.722	0.676	0.745	0.202	0.43	0.29	0.123
depresjon	XGBoost	record-wise	0.802	0.776	0.872	0.137	0.82	0.07	0.039
psykose	Majority class	subject-wise	0.661	0.398	0.476	0.224	-3.95	0.00	0.000
psykose	Majority class	record-wise	0.661	0.398	0.499	0.224	0.31	0.00	0.000
psykose	Logistic regression	subject-wise	0.856	0.842	0.906	0.115	0.90	-0.60	0.086
psykose	Logistic regression	record-wise	0.868	0.855	0.926	0.103	1.07	-0.62	0.078
psykose	Random forest	subject-wise	0.850	0.829	0.889	0.118	0.95	-0.12	0.048
psykose	Random forest	record-wise	0.878	0.861	0.936	0.093	1.32	-0.17	0.048
psykose	XGBoost	subject-wise	0.862	0.842	0.899	0.108	0.71	0.15	0.046
psykose	XGBoost	record-wise	0.891	0.876	0.942	0.082	0.91	0.07	0.026
hyperaktiv	Majority class	subject-wise	0.504	0.335	0.496	0.250	0.00	0.00	0.000
hyperaktiv	Majority class	record-wise	0.504	0.335	0.496	0.250	0.00	0.00	0.000
hyperaktiv	Logistic regression	subject-wise	0.461	0.460	0.463	0.286	-0.21	-0.01	0.160
hyperaktiv	Logistic regression	record-wise	0.543	0.543	0.577	0.252	0.21	0.01	0.074
hyperaktiv	Random forest	subject-wise	0.463	0.462	0.447	0.286	-0.32	0.00	0.156
hyperaktiv	Random forest	record-wise	0.571	0.571	0.603	0.245	0.57	0.00	0.055
hyperaktiv	XGBoost	subject-wise	0.460	0.460	0.449	0.330	-0.15	0.00	0.248
hyperaktiv	XGBoost	record-wise	0.557	0.557	0.586	0.271	0.26	0.00	0.151

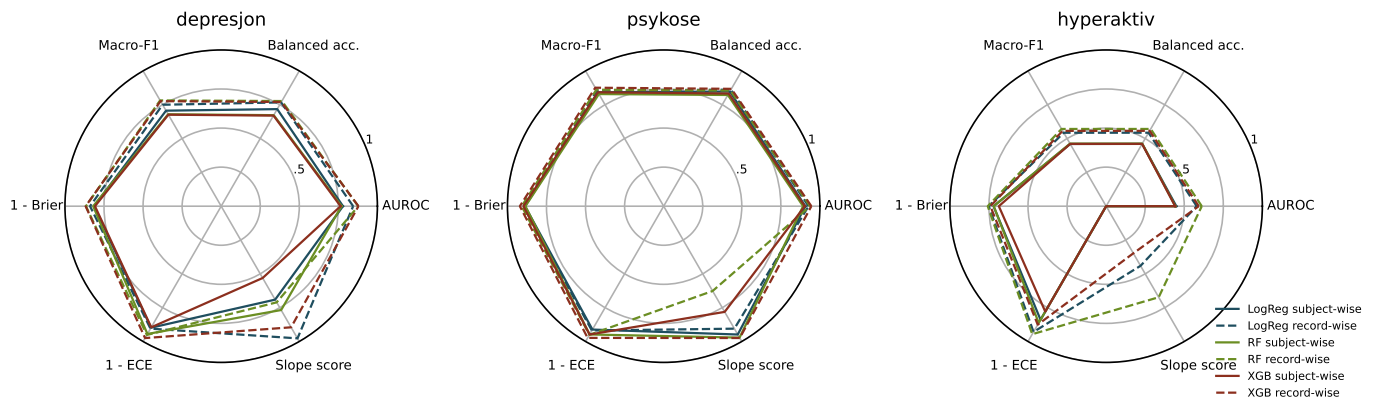


Fig. S4. Multi-metric profile of the models per cohort under subject-wise (solid) and record-wise (dashed) cross-validation.